

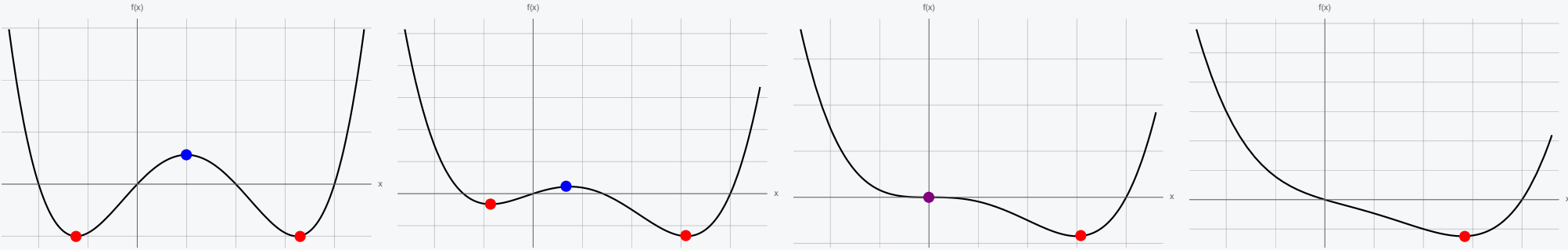


Local Floer Chain Complex across Hamiltonian Bifurcation (Ongoing work with H.Jo)

Dongho Lee, QSMS, Seoul National University
Presented at **Geometry Space Surrey**, London, UK, 10 June 2026



Toy Example : Function



This is graph of $f_\varepsilon(x) = x(x^2 + t)(x - 1)$ for $\varepsilon = -1, -1/2, 0$ and 1 .

See the change of critical points.

- $\varepsilon < 0$, two minima (●) and one maximum. (●)
- $\varepsilon = 0$, one minimum and one maximum meets at the inflection point. (●)
- $\varepsilon > 0$, one minimum survives. (●)

This phenomnom also happens for **periodic orbits**.

Bifurcation of Hamiltonian Periodic Orbits

Let $H_\varepsilon : W \rightarrow \mathbb{R}$ be a 1-parameter family of Hamiltonians and γ_0 be a periodic orbit of H_0 . **Floquet multiplier** of γ_0 is the eigenvalue λ of the linearized return map.

We call γ_0 is **non-degenerate** if $\lambda \neq 1$.

If γ_0 is non-degenerate, there exists an **orbit cylinder** for small ε ,

$$\Gamma = \{(\gamma_\varepsilon, \varepsilon) : \gamma_\varepsilon \text{ is a periodic orbit of } H_\mu\} \simeq S^1 \times I.$$

If γ_0 is degenerate, the orbit cylinder breaks and a **bifurcation** happens:

- The stability type (elliptic / hyperbolic) of the orbit might change.
- The family γ_ε might vanish, or another orbits might be born.
- If $\lambda \neq 1$ but $\lambda^k = 1$, then k -th cover of the orbit bifurcates.

Conley-Zehnder Index and Floer Homology

The **Conley-Zehnder index** $\mu(\gamma)$ is an integer associated to the periodic orbit. ([CZ83]) CZ index of a family γ_ε changes *only if* it degenerates.

$\Rightarrow$ We can track bifurcation by computing $\mu(\gamma_\varepsilon)$. (cf. [AFvK⁺], [AB25])

CZ index gives a grading of **Floer homology**, which is a topological invariant. (cf. [Flo89])

- Generators are periodic orbits, graded by CZ index
- Differential is given by *Floer cylinders*, a solution of an elliptic PDE with boundary asymptotic to the periodic orbits.

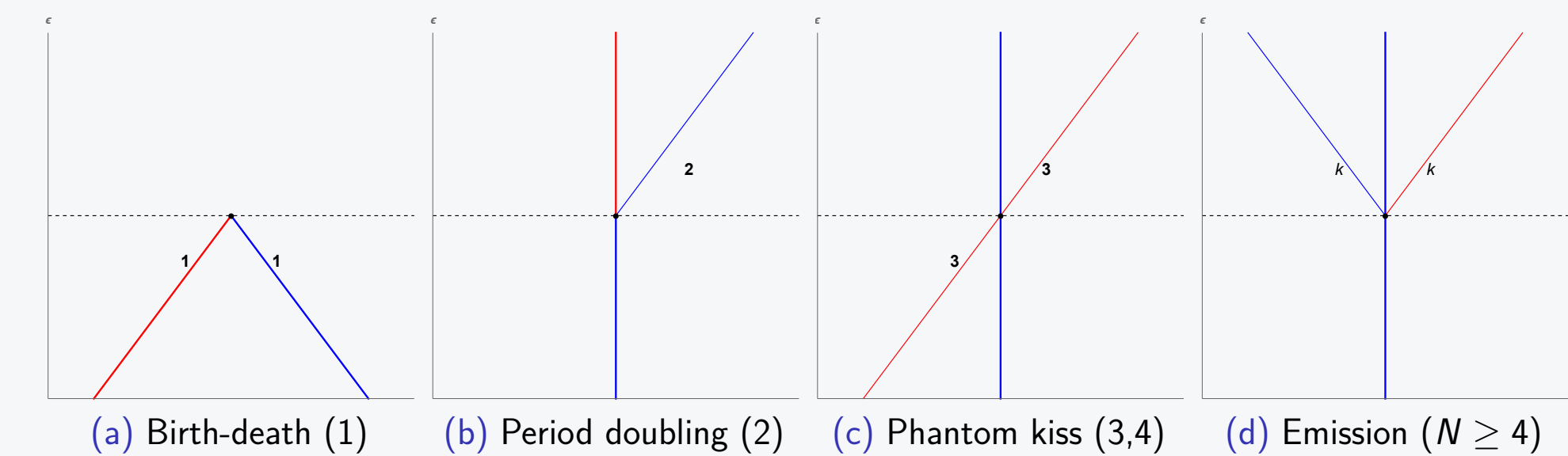
Bypassing a bifurcation, a number of periodic orbits and their CZ index changes, but the homology determined by them is invariant.

Slogan. Bifurcation gives a mutation of Floer chain complex.

Meyer's Classification of Generic Hamiltonian Bifurcation

Generic Hamiltonian bifurcations can be classified into 5 types. ([Mey70])

Those are classified by the covering number of the orbit: $N = 1, 2, 3$ and $N \geq 4$.



Lines are orbit cylinders and numbers are approximate periods of the orbits.

Blue is elliptic and red is hyperbolic.

The fifth type, called *murder*, is same as period doubling with opposite stability type.

There are other types if there is a symmetry (e.g. CRTBP), but all based on these types.

Normal Form and Gradient Revolution

Goal. Construct an **explicit Floer chain complex** near bifurcations.

Idea. The structure of a gradient flow of the Hamiltonian (of the return map) on the Poincaré section determines the structure of the local Floer chain complex.

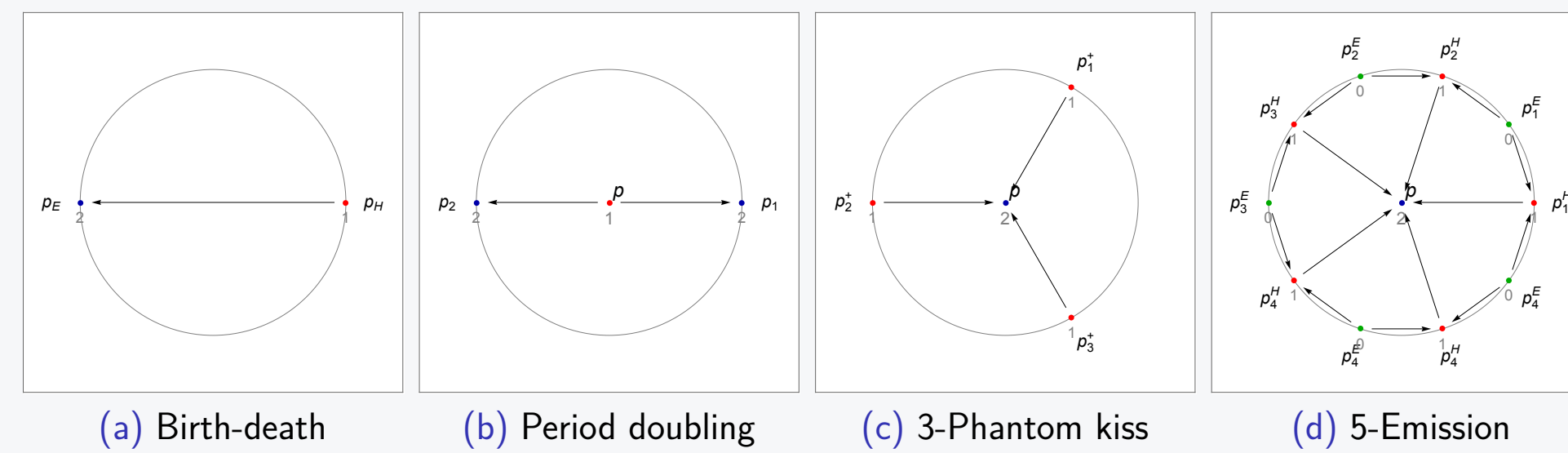
1. For k -cover bifurcation, we can describe the k -th iterate of return map as a time 1-flow of a Hamiltonian G_μ .
2. Using Birkhoff normal form (cf. [MH92]), we can write the Hamiltonian by $\mathbb{Z}_k$ -symmetric polynomials up to finite order, i.e. $G_\mu \in \mathbb{R}[r^2, r^k \cos k\theta, r^k \sin k\theta]$
3. Critical points of G_μ corresponds to the periodic orbit.
Moreover, the Morse index determines the stability type and CZ index.
4. Define **model Hamiltonian** on a (covering space of) neighborhood of the orbit by

$$H(\rho, \varphi, u, v) = h(\rho) + G_\mu(u, v).$$

This contains the information of the return map and CZ index.

5. Take a Morse trajectory of G_μ and rotating it along φ , we get a Floer cylinder, which we call a **gradient revolution**. The construction is equivariant to the covering map.

Structures of Gradient Flows



The figures describe the rigid positive gradient flows on Poincaré section.

Opposite direction is also possible; here we only describe one type.

Points with same color indicates the same orbit. (k -points = k -periodic orbit.)

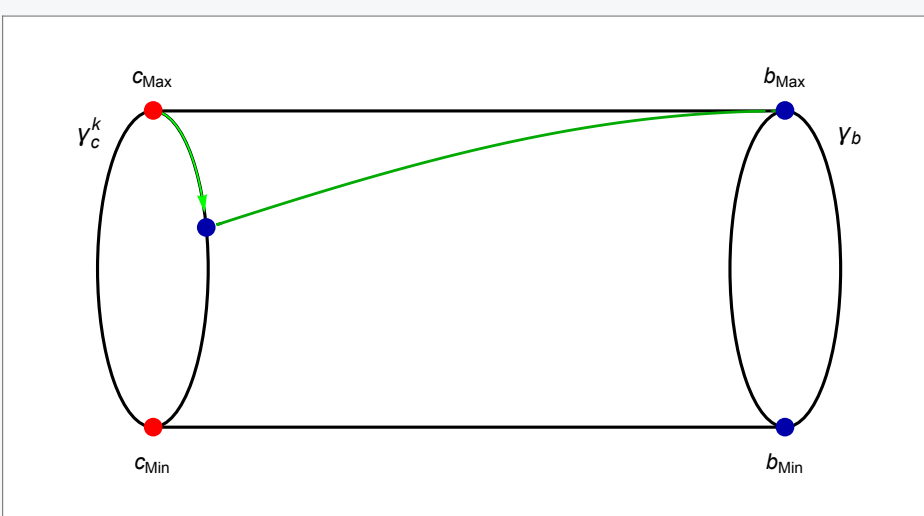
Morse-Bott Cascade Counting

The **Morse-Bott cascade** is used to count the Floer differential.

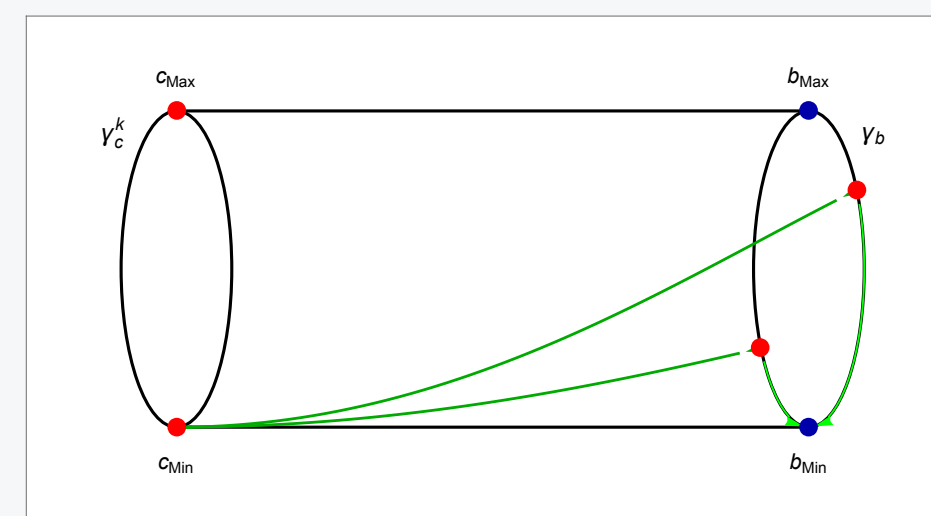
Key Lemma

If there exists a gradient revolution from a k -th cover γ_c^k to a simple orbit γ_b with index difference 1, the (un-signed) Morse-Bott cascade counting gives $\partial_F(\hat{\gamma}_c^k) = \hat{\gamma}_b$, $\partial_F(\check{\gamma}_c^k) = k\check{\gamma}_b$.
If the gradient revolution has the opposite direction, we have $\partial_F(\hat{\gamma}_b) = k\hat{\gamma}_c$, $\partial_F(\check{\gamma}_b) = \check{\gamma}_c$.

Sketch of proof (For those who know Morse-Bott cascade).



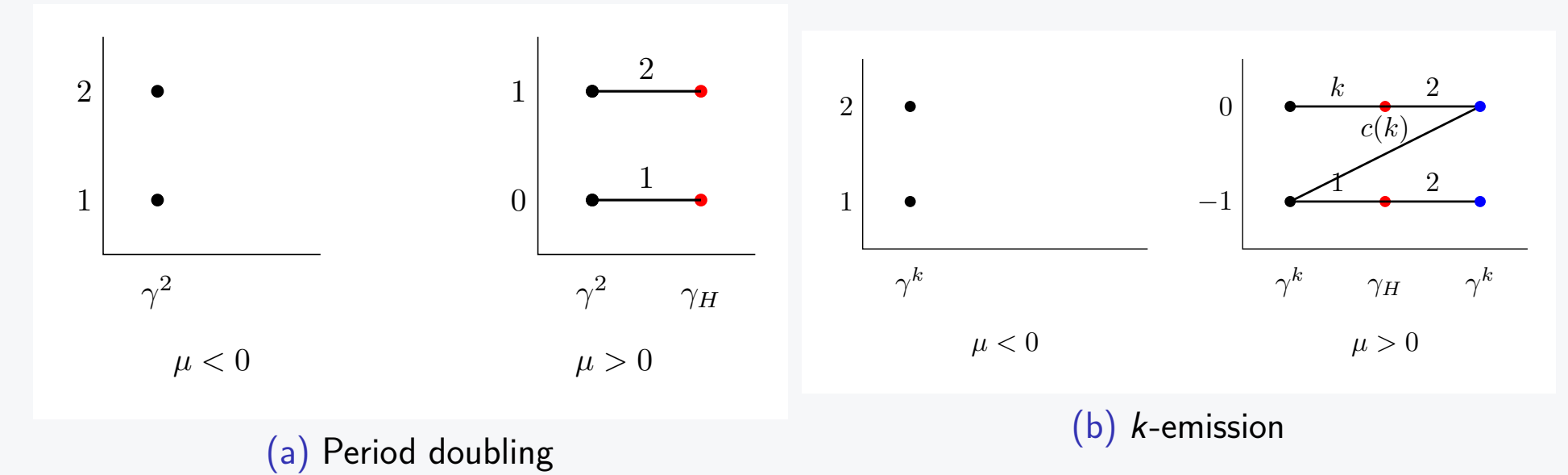
- $\hat{\gamma}_c^k \rightarrow \hat{\gamma}_b$ consists of Morse trajectory from $\gamma_{c,\max}$ to $ev^{-1}(\gamma_{b,\min})$.
- $ev^{-1}(\gamma_{b,\min})$ is only 1 point, so there is one Morse-Bott cascade.



- $\check{\gamma}_c^k \rightarrow \check{\gamma}_b$ consists of Morse trajectories from $ev(\gamma_{c,\min})$ to $\gamma_{b,\min}$.
- Since γ_c^k is k -th cover, the image of the evaluation map consists of k distinct points.

Current Results and Further Directions

- The resulting chain complex satisfies the invariance of local Floer homology. This provides the basic example of Floer chain complex mutation.



- We also get the profile of change of CZ index for each bifurcation type.
- The method can directly applied to systems with symmetries, for example CRTBP, Hill's lunar problem and Hénon-Heiles system.

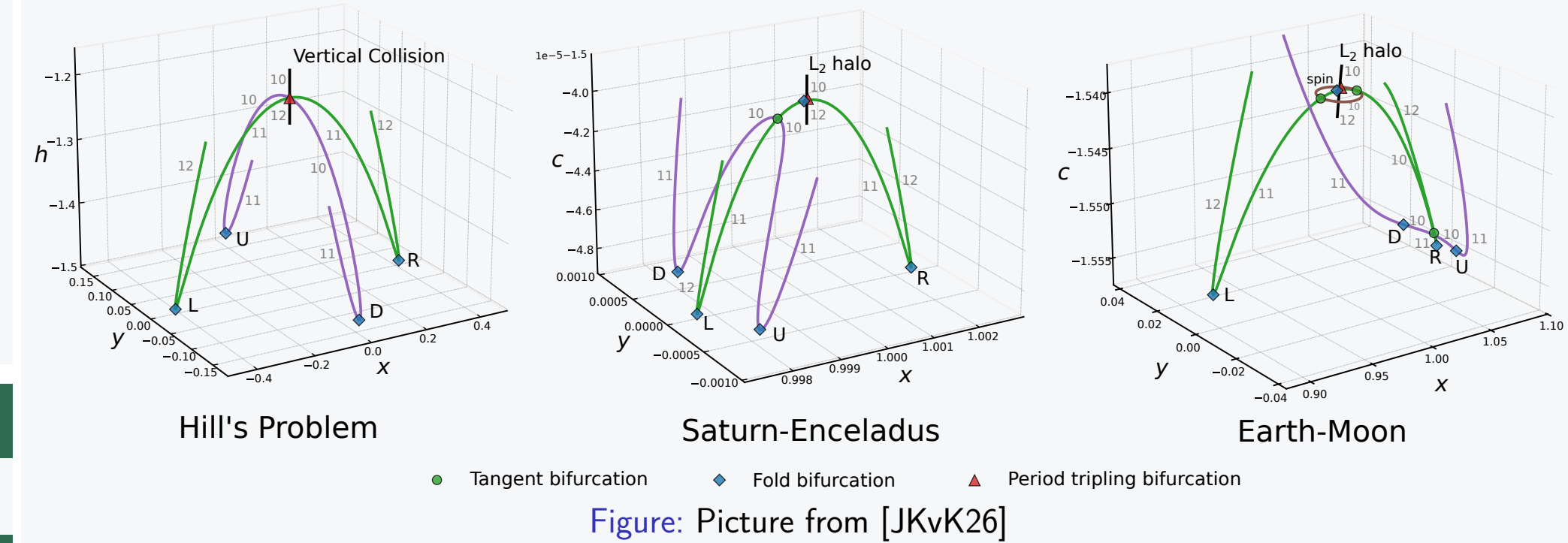
Application to the Restricted Three-body Problem

There are many complicated bifurcations in systems like CRTBP.

For example, there is a *double emission* occuring at the triple cover of the vertical collision orbit in the Hill's lunar problem.

As μ turns on, this breaks into 3 bifurcations: phantom kiss, birth-death and pitchfork.

With our machinery, such phenomena can be analyzed systematically.



In the on-going work with Chankyu Joong, we are analyzing this symmetry-breaking phenomena in terms of Floer homology and Conley-Zehnder indices.

References

- [AB25] Cengiz Aydin and Alexander Batkhin. Studying network of symmetric periodic orbit families of the Hill problem via symplectic invariants. *Celestial Mech. Dynam. Astronom.*, 137(2):Paper No. 12, 77, 2025.
- [AFvK⁺] Cengiz Aydin, Urs Frauenfelder, Otto van Koert, Dayoung Koh, and Agustin Moreno. Symplectic geometry and space mission design.
- [CZ83] C. C. Conley and E. Zehnder. The Birkhoff-Lewis fixed point theorem and a conjecture of V. I. Arnold. *Invent. Math.*, 73(1):33–49, 1983.
- [Flo89] Andreas Floer. Symplectic fixed points and holomorphic spheres. *Comm. Math. Phys.*, 120(4):575–611, 1989.
- [JKvK26] Chankyu Joong, Dayung Koh, and Otto van Koert. Bifurcations of highly inclined near halo orbits using moser regularization, 2026.
- [Mey70] K. R. Meyer. Generic bifurcation of periodic points. *Transactions of the American Mathematical Society*, 149(1):95–107, 1970.
- [MH92] Kenneth R. Meyer and Glen R. Hall. *Introduction to Hamiltonian dynamical systems and the N-body problem*, volume 90 of *Applied Mathematical Sciences*. Springer-Verlag, New York, 1992.

